



METU - Department of Computer Engineering
CENG 7880 – Trustworthy and Responsible AI
2026-2027 Fall



Webpage: <https://metu-trai.github.io>

Forum: ODTUclass page of the course.

Instructor: Sinan Kalkan, skalkan@metu.edu.tr (Office hours: by appointment)

Lectures: Mon: 9:40-12:30 [BMB-5]

Credits: METU: 3 Theoretical, 0 Laboratory; ECTS: 8.0

Catalog Description: Ethical concepts and principles of trust and responsibility in AI; computational methods for dependability and robustness in AI; computational methods for explainability in AI; computational methods for bias and fairness in AI.

Background Requirements: Background in deep learning is a must. The students must have taken CENG403 (Introduction to Deep Learning) or CENG501 (Deep Learning).

Reference Material: (1) Lu, Q., Zhu, L., Whittle, J., & Xu, X. (2023). Responsible AI: Best practices for creating trustworthy AI systems. Addison-Wesley Professional. (2) Bartneck, C., Lütge, C., Wagner, A., & Welsh, S. (2021). An introduction to ethics in robotics and AI (p. 117). Springer Nature.

Grading:

Quizzes	20%
Project	35%
Final Exam	45%

Warnings

Participating in the final exam is subject to the following conditions: (i) Attending 60% of the quizzes. (ii) Completing half of the project. Not satisfying these will result in NA as the grade.

Tentative Schedule:

Week & Date		Topic
1	28 Sep	Course logistics and overview [Intro to trust and responsibility in AI: Social, ethical and policy implications of AI]
2	5 Oct	Overview [Deep/machine learning concepts (fundamentals)]
3	12 Oct	Overview [Deep/machine learning concepts (recent trends)]
4	19 Oct	Robust AI [Fundamentals of Robust AI]
5	26 Oct	Robust AI [Adversarial Attacks & Corruptions]
6	2 Nov	Robust AI [Defense Mechanisms and Robust Training]
7	9 Nov	Robust AI [Quantifying Uncertainty and Calibration of Deep Networks]
8	16 Nov	Explainable AI [Fundamentals of Explainable AI]
9	23 Nov	Explainable AI [Model-specific Explainability Methods: Explainable shallow models, prototype-based models, concept bottleneck models, attention mechanisms for explainability]
10	30 Nov	Explainable AI [Post-Hoc Explanation Techniques: CAM, GradCAM, feature-based saliency estimation, counterfactual reasoning]
11	7 Dec	Explainable AI [Evaluating explainability, methods and metrics]
12	14 Dec	Bias and Fairness in AI [Sources of bias; definitions of fairness; measures of fairness]
13	21 Dec	Bias and Fairness in AI [Data-level and in-processing methods for fairness]
14	28 Dec	Bias and Fairness in AI [Regularization-based and post-processing methods for fairness]

Professionalism and Ethics

Students are expected to complete the course activities on their own, unless otherwise stated. Sharing your work with others, uploading the course material to online websites for seeking solutions, and/or presenting someone else's work/solution as your own work will be considered cheating. For a comprehensive cheating definition, please refer to:

<https://oidb.metu.edu.tr/sites/oidb.metu.edu.tr/files/Academic%20Integrity%20Guide%20for%20Students.pdf>

When a breach of the code of ethics occurs (cheating, plagiarism, deception, etc.), the student will be added to the BLACK list, and the instructor can apply one of the following:

1. Perform an oral or written test for the students to confirm their knowledge and assess their grades. If the follow-up exam (oral/written) results in failure, the instructor can apply one of the other items below.
2. Give a "zero" grade directly for the relevant exam/assignment,
3. Give a "zero" grade for a larger part or all of the assignments,
4. Give a failing letter grade for the course,
5. Forward the case to the discipline committee.

Makeup Exam Policy

There will be no makeup exams unless the student has a valid medical leave approved by the METU Health Center. Students who provide a valid medical leave approved by the METU Health Center will be eligible for a makeup examination, subject to the following regulations:

1. There will be a single makeup exam.
2. The makeup examination will be conducted after the final exams at the end of the semester.
3. The makeup examination will be comprehensive and can cover all topics addressed throughout the semester, regardless of the specific exam missed (e.g., quizzes, midterm, final).
4. The grade obtained in the comprehensive makeup examination will replace the grade of the missed assessment.
5. In the unlikely cases where a student missed multiple exams with valid medical documentation, the grade of the makeup exam will replace the grades of all the missed elements.